

STATEMENT OF WORK (SOW)

Device Risk Analysis | Case Study No. 86

Candidate: Atharva Gahine | B.Tech CSE Semester V (2024–2028)

1. Project Scope & Deliverables

This Statement of Work (SOW) defines the contractual and academic scope for developing the Device Risk Analysis system. The project delivers raw and processed datasets, an executed 22-section Jupyter Notebook, serialized champion ML models, an interactive Streamlit application, and seven complete academic documentation artifacts.

2. Milestone Timeline

Milestone	Phase Description	Timeline	Status
M1	Problem Formulation & Telemetry Collection	Week 1	COMPLETED
M2	Data Cleaning, Preprocessing & EDA	Week 2	COMPLETED
M3	Algorithm Benchmark & Tuning	Week 3	COMPLETED
M4	Streamlit Dashboard Development	Week 4	COMPLETED
M5	System Verification & QA Testing	Week 5	COMPLETED
M6	Academic Documentation & Final Review	Week 6	COMPLETED

3. Risk Management & Acceptance Criteria

Data Leakage Prevention: Transformers fit strictly on training folds inside scikit-learn Pipeline objects.

Performance Benchmark: Champion model must achieve $\geq 85.0\%$ accuracy (Actual: 89.00% Accuracy, 0.8898 F1, 0.9756 ROC-AUC).